

CD-A

(Differential) TDR using TAUPE Sensor Cables

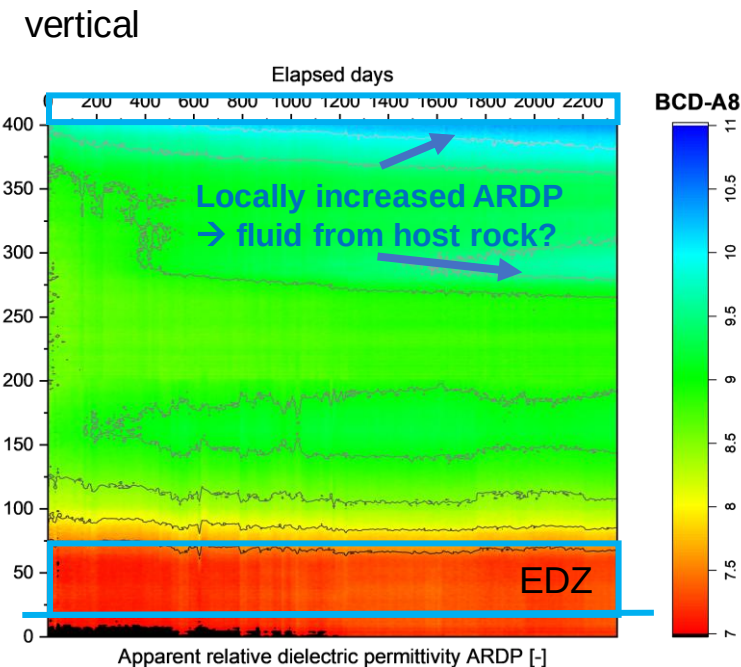
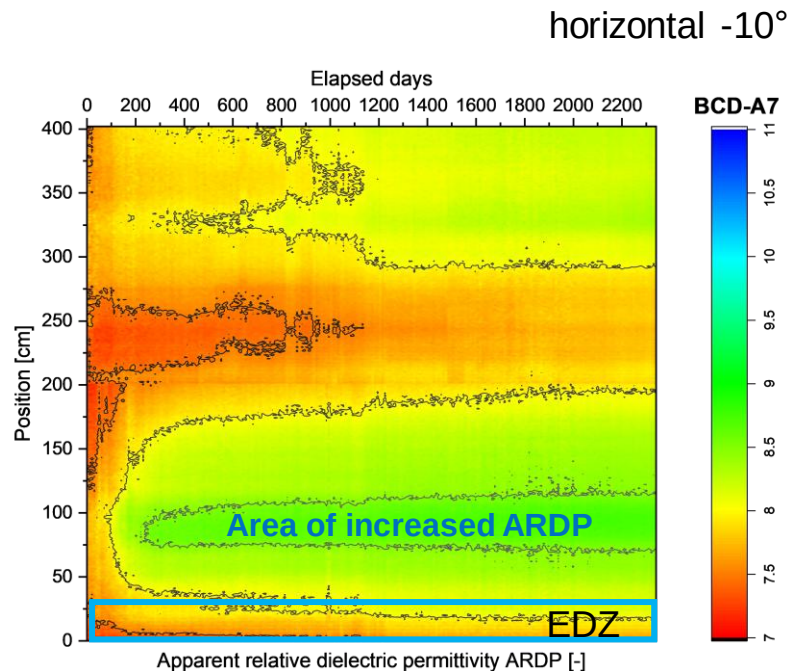
F. Königer, R. Schuhmann, G. Ziefle (BGR), D. Jaeggi (MT)

ISU GmbH – Ingenieur-Gesellschaft für Sensorik in der Umwelttechnik mbH



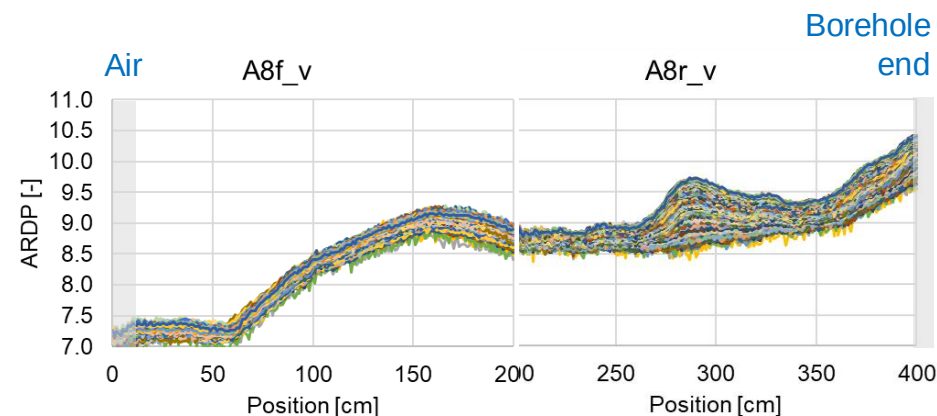
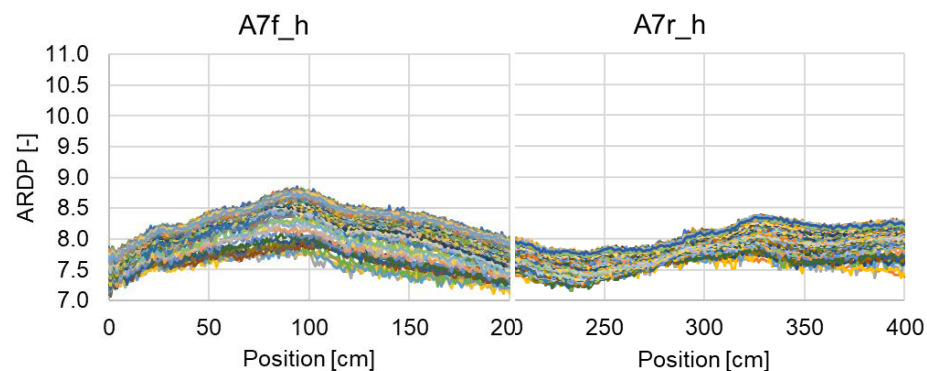
Closed Twin: ARDP

No significant changes
in ARDP during more
than six years



No significant changes
in ARDP, areas with
locally increased ARDP

Remark:
Taupe tube sensor was
not completely inserted,
bore hole starts at 15 cm



Start: 22.10.2019

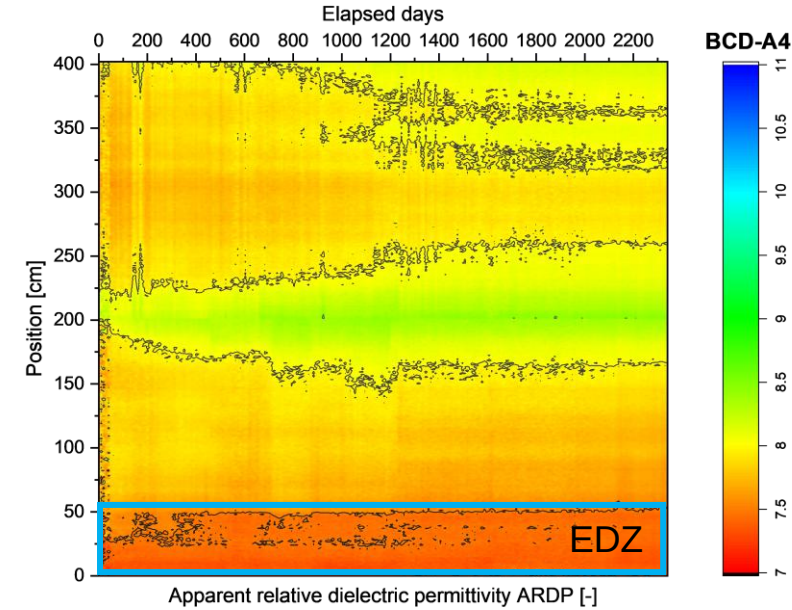
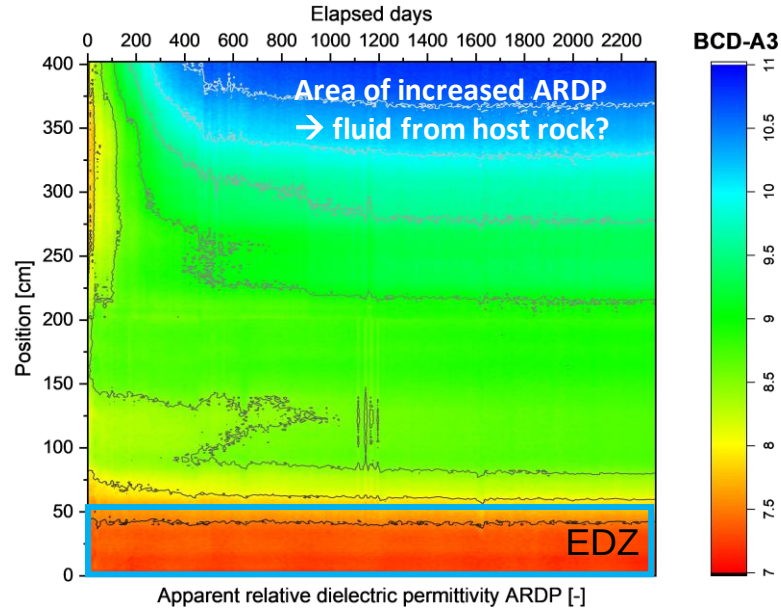
Present : 17.04.2026 = 2361 days

Open Twin: ARDP

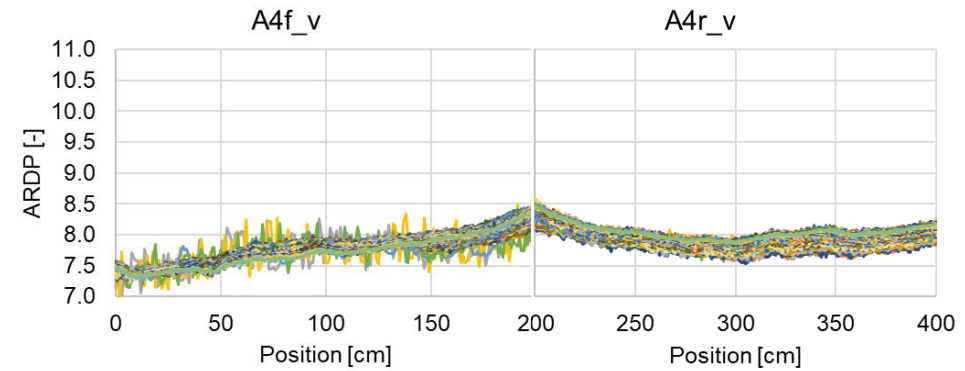
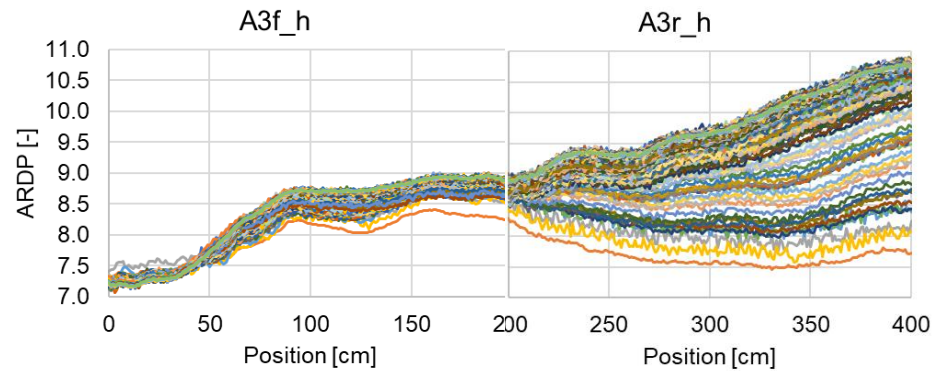
horizontal -10°

vertical

No significant changes
in ARDP in first 2 m,
areas with locally
increased ARDP at
the end of the bore hole



No significant changes
in ARDP during more
than six years



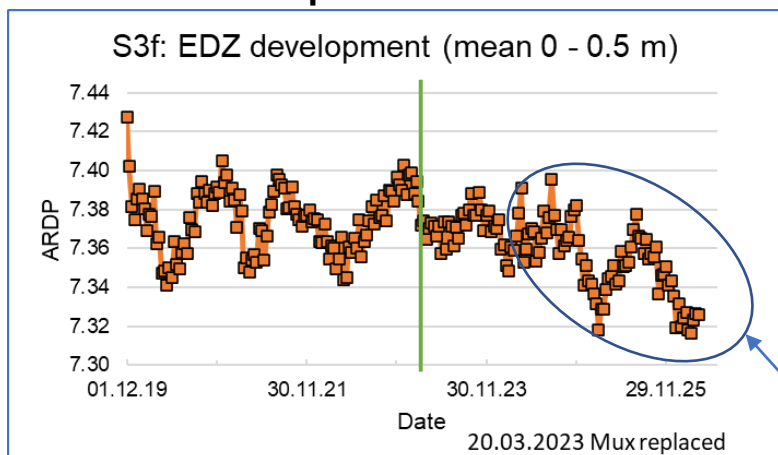
Start: 22.10.2019

Present : 17.04.2026 = 2361 days

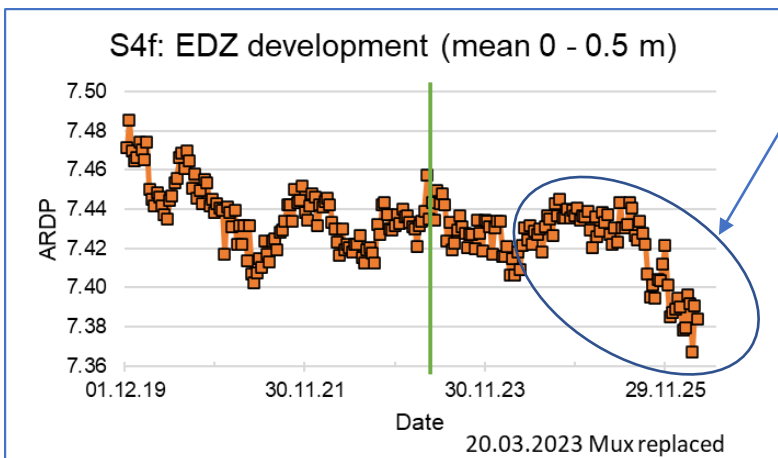
Meteorological effects on EDZ

(Mean amplitude over first 0.5 m)

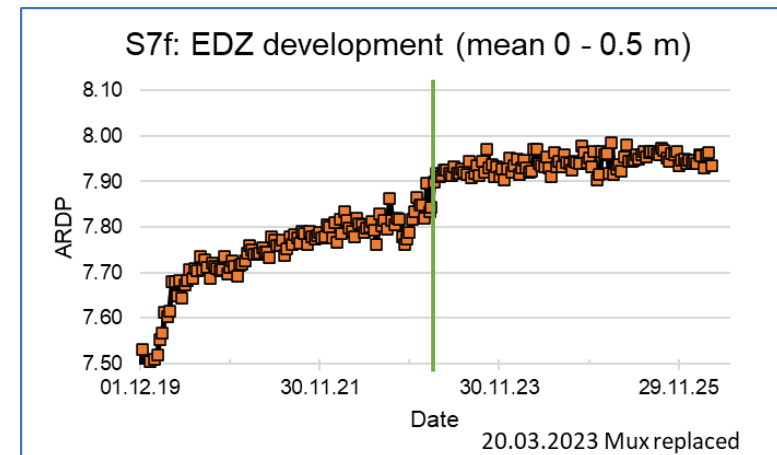
Open Twin



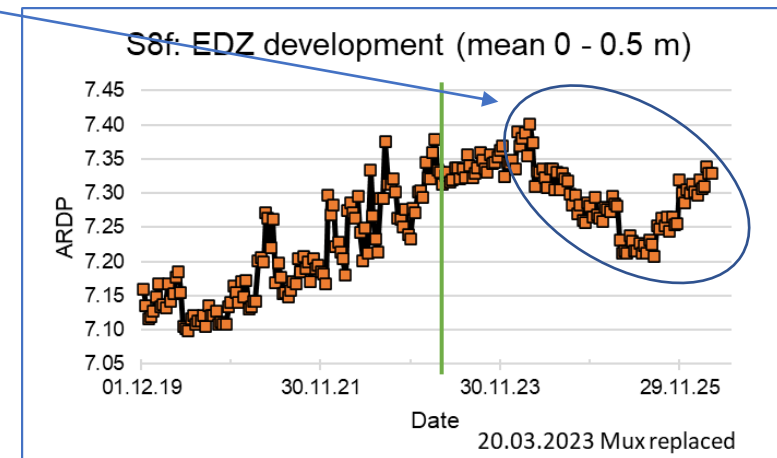
Major variations on ARDP due to meteorological effects, stronger in the horizontal direction



Closed Twin



No visible meteorological effects in both directions



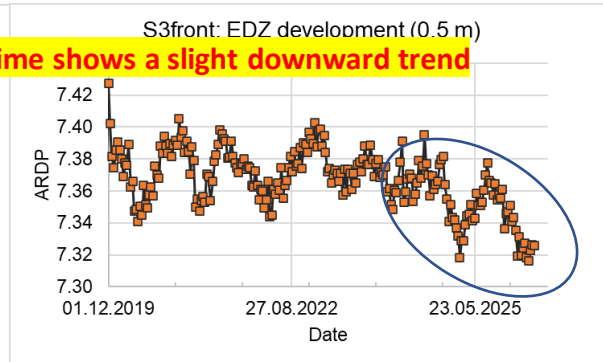
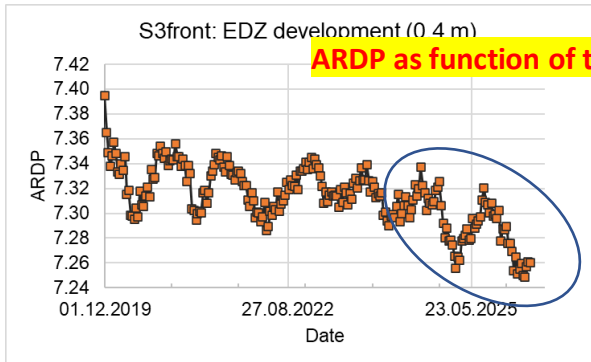
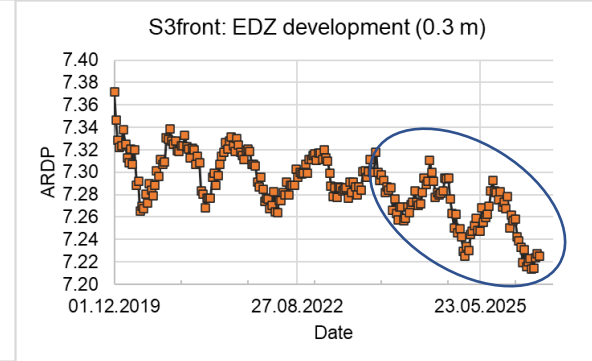
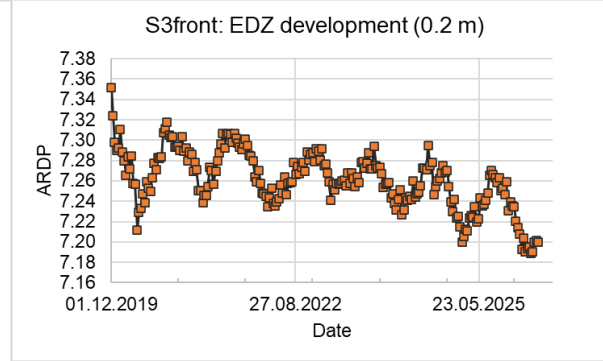
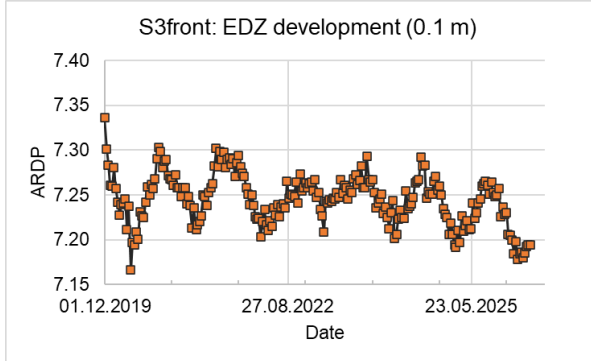
A significant reduction in the ARDP from the start of 2025 appears (not at S7front)

Start: 22.10.2019

Present : 17.04.2026 = 2361 days

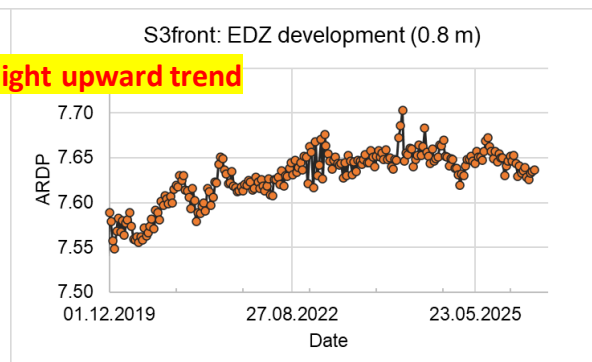
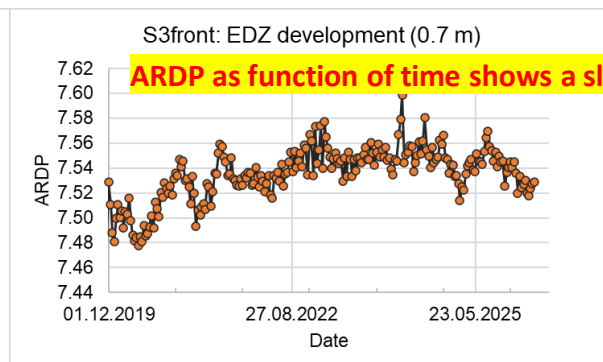
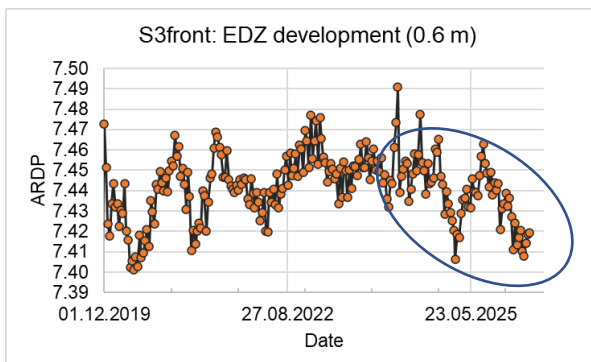
Meteorological effects on EDZ at S3front

(Mean amplitude over varying depth, starting at 0 m)



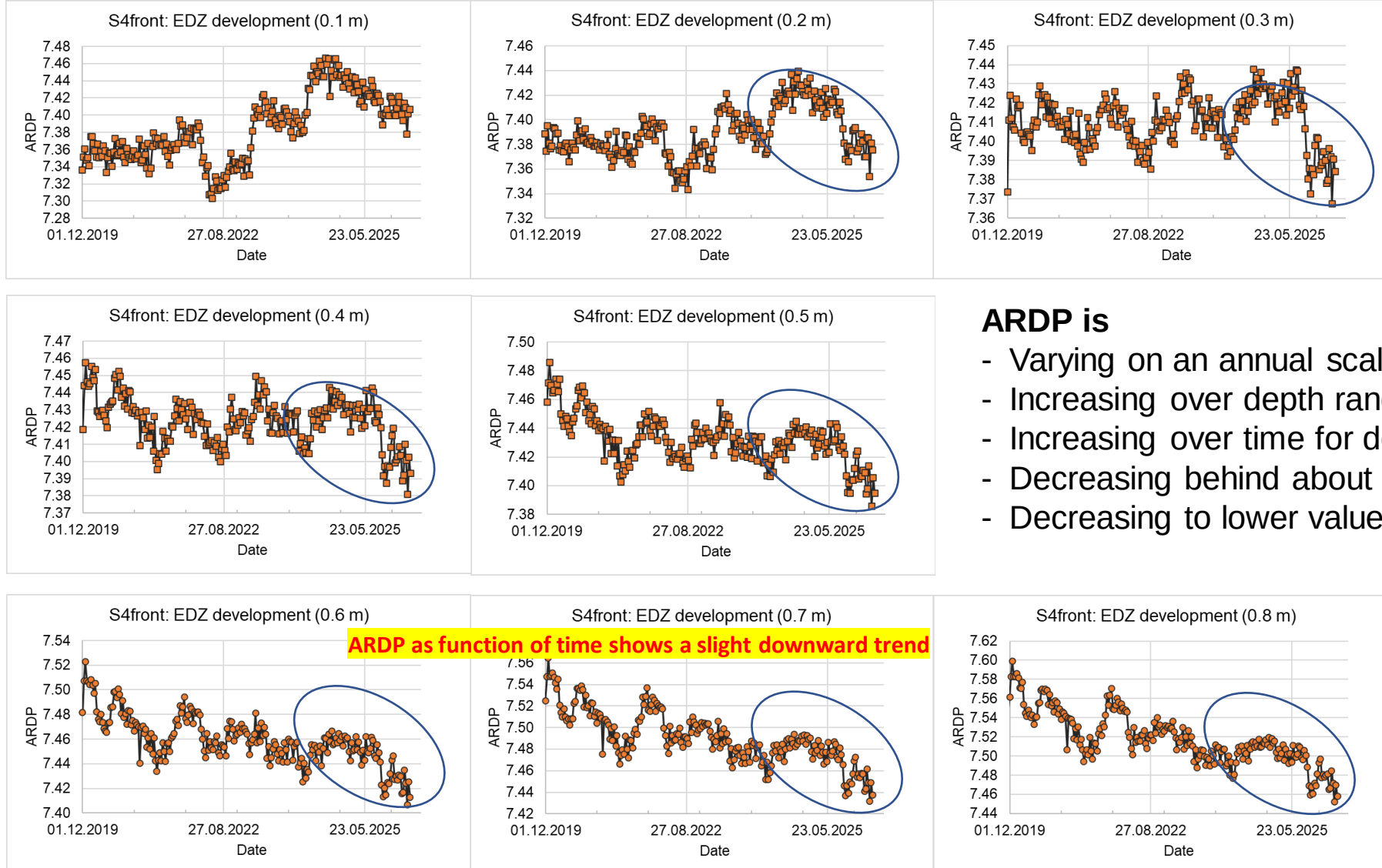
ARDP is

- Varying on an annual scale
- Increasing over depth range
- Decreasing over time for depth behind 0.6 m
- Increasing behind about 0.6 m
- Decreasing to lower values since start of 2025, not for ranges behind 0.7 m



Meteorological influences on EDZ at S4front

(Mean amplitude over varying depth, starting at 0 m)



ARDP is

- Varying on an annual scale
- Increasing over depth range
- Increasing over time for depth below 0.4 m
- Decreasing behind about 0.4 m
- Decreasing to lower values since start of 2025

Summary

Closed Twin

Horizontal -10°

- EDZ narrow
- Moderate ARDP in most parts
- Medium ARDP around 40 – 110 cm depth, slightly increasing
- Starting increased ARDP zone at rear end

Vertical

- EDZ quite pronounced
- Medium ARDP, in most parts only slightly increasing
- **Higher ARDP** around 150 cm and at borehole end (significant)

Open Twin

Horizontal -10°

- EDZ pronounced
- Slight variations of ARDP between 80 and 300 cm
- **Significantly higher ARDP** behind about 300 cm (Borehole end)
Potentially: Wet spots? Perturbation/fault?

Vertical

- EDZ quite pronounced
- Low/medium ARDP over complete borehole
- Only slight variations of ARDP over time
- Slightly increased ARDP zone at sensor connection -> artificial ?